ST MARY'S DSG, KLOOF

GRADE: 10 18 February 2021

MATHEMATICS TEST
Products, Factors, Fractions

TIME: 40 min TOTAL: 37 MARKS

Name: _____

QUESTION 1:

Expand the following and simplify fully:

a) $(x+6)(x^2-6x+36)$ (2)

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b) $-(x-2)^2$ (3)

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c) $\left(3x-\frac{1}{x}\right)\left(3x+\frac{1}{x}\right)$ (3)

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[8]

QUESTION 2:

Factorise completely:

a) $x^2 - 7x - 18$ (2)

b) $27y^3 - 8$ (3)

c) $x^4 - y^4$ (3)

d) $(x + y)^2 + 8(x + y) + 15$ (3)

e) $b^2 - c^2 - b - c$ (4)

[15]

QUESTION 3:

Simplify the following fractions:

a) $\frac{x^2+2x}{x^2-x-6} \times \frac{x-3}{x}$ (3)

b) $\frac{9-3a}{a^2-4} \div \frac{3a^2-11a+6}{a+2}$ (5)

[8]

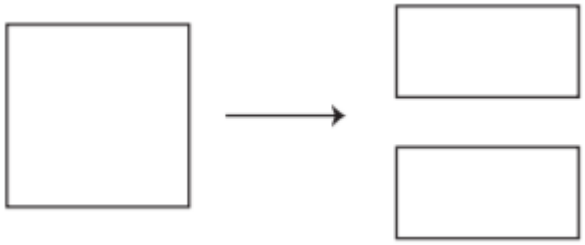
QUESTION 4:

a) If $x^2 + 2x - 5 = (x + 1)^2 + k$, find the value of k. (2)

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b) A square is cut into two identical rectangles, as shown. Each of these two rectangles has perimeter 24 cm. What is the area of the original square? **WORKING MUST BE SHOWN**. (4)



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[6]